Peierls-Transport Obstruction and the Minimal Many-Body Escape

Exact multi-block one-body no-go theorem and a fully covariant local null-row model

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Review-integrated revision, July 26, 2026

Abstract

We prove a sharp obstruction to composition-space hopping in exact multi-block projected-Hubbard and quantum-geometric-nesting parents. If the complete zero manifold contains the separate one-pair antisymmetrized-geminal-power states, then every number-conserving one-body null row is block preserving already at zero twist. A complete covering-lattice Peierls connection transports the isolated interaction singular subbundles by one common analytic unitary. The resulting commutator source lies in the range of the unperturbed null map and is removed exactly by the frustration-free least-squares projection. The surviving source preserves every block charge, so the exact composition-space curvature operator has no off-diagonal entries. Fixed finite range or uniform exponential locality removes the only finite-torus alias loophole. The blocks may be interaction subbundles rather than normal-state energy bands.

We then give a strictly nearest-neighbor, translation-invariant, positive-square many-body null-row model with two extensive inequivalent blocks, a complete product-AGP zero manifold in every even sector, a termwise covariant Peierls lift, and an exact non-diagonal one-pair composition operator

$$Q^{(1)} = \begin{pmatrix} 2j_1 + 4g^2 & -4g^2 \\ -4g^2 & 2j_2 + 4g^2 \end{pmatrix}.$$

The escape is minimal at the level of theorem hypotheses: every assumption is retained except that the transfer-generating null row is multiplicative and many body rather than additive and one body.

1 Introduction

Quantum-geometric nesting (QGN) and projected positive-semidefinite Hubbard constructions provide frustration-free superconducting parents whose paired zero states and flat-connection curvatures can be computed exactly [1]. With one interaction singular block, the fixed-number zero state is a single antisymmetrized geminal power (AGP). With two or more inequivalent extensive singular subbundles, the zero manifold instead contains one product AGP for every allowed distribution of pairs among the blocks [2].

This degeneracy makes the action of electromagnetic flux qualitatively important. If the physical twist preserves each block charge, the fixed-total-number response is only the lower envelope of independent composition branches. A genuine transfer source would instead produce an operator on the composition lattice: neighboring compositions would hybridize, and the physical curvature would be the smallest eigenvalue of that matrix rather than the minimum of its diagonal entries. Finding a local microscopic realization of this composition hopping was therefore the final unresolved step in the two-block program.

Two natural one-body candidates failed for covariance reasons. A smooth common hybridization of the blocks produces only a removable frame-transport commutator, while a band-filtered multi-site bridge remains a zero row when the bridge kernel and the projectors are Peierls twisted consistently [3]. The purpose of this note is to decide whether those failures were accidental. We prove that they are structural throughout the exact local one-body class, including interaction-defined blocks that need not coincide with normal-state energy bands. We then construct a strictly range-one many-body null row that keeps every other desired hypothesis and produces an exact non-diagonal composition response.

2 Multi-block zero manifold and curvature map

Let

$$\mathfrak{h} = \bigoplus_{a=1}^{q} \mathfrak{h}_a, \qquad \dim \mathfrak{h}_a = 2L_a, \tag{1}$$

and let each block carry a skew-unitary pairing form J_a. Define

$$\eta_a^+ = \frac{1}{2} c^\dagger J_a c^\dagger, \qquad |\mathbf{n}\rangle = \bigotimes_{a=1}^{q} \left[\frac{(L_a - n_a)!}{n_a! L_a!}\right]^{1/2} (\eta_a^+)^{n_a} |0\rangle. \tag{2}$$

At fixed total pair number n, put

$$\mathcal{C}_n = \{\mathbf{n} : 0 \le n_a \le L_a,\ \sum_a n_a = n\}, \qquad \mathcal{Z}_n = \operatorname{span}\{|\mathbf{n}\rangle : \mathbf{n} \in \mathcal{C}_n\}. \tag{3}$$

Consider an analytic positive-square family

$$H(A) = \frac{1}{2} D(A)^\dagger D(A) = \frac{1}{2} \sum_\lambda S_\lambda(A)^\dagger S_\lambda(A). \tag{4}$$

Assume that $\mathcal{Z}_n = \ker D(0)$ in the sector under consideration and that the zero cluster is separated from the positive spectrum at the fixed finite size. For a real twist direction v, define

$$\mathcal{B}_v|\psi\rangle = \big(\partial_v S_\lambda(0)|\psi\rangle\big)_\lambda, \qquad \Pi = P_{\ker D(0)^\dagger}. \tag{5}$$

Degenerate frustration-free perturbation theory gives the exact second-order operator

$$Q_v^{(n)} = T_v^\dagger T_v, \qquad T_v = \Pi \mathcal{B}_v|_{\mathcal{Z}_n}. \tag{6}$$

Thus composition hopping is present precisely when projected source vectors from two different product-AGP compositions have nonzero overlap.

For comparison with the multi-block reduction, write

$$b_{\lambda v}^{ab} = P_a\, \partial_v s_\lambda(0)\, P_b, \qquad \boldsymbol{\zeta}_v^{a\leftarrow b} = \big(d\Gamma(b_{\lambda v}^{ab})|\mathbf{e}_b\rangle\big)_\lambda \quad (a \neq b). \tag{7}$$

Let Π_{ab} denote the restriction of the target-kernel projector to the sector with one fermion in each of blocks a and b. The reduced transfer norm and opposite-direction overlap are

$$k_v^{a\leftarrow b} = \|\Pi_{ab} \boldsymbol{\zeta}_v^{a\leftarrow b}\|^2, \qquad \omega_v^{ab} = \left\langle \Pi_{ab} \boldsymbol{\zeta}_v^{b\leftarrow a}, \Pi_{ab} \boldsymbol{\zeta}_v^{a\leftarrow b} \right\rangle. \tag{8}$$

For two blocks at fixed total filling, ω_v^{ab} multiplies the only possible off-diagonal matrix element between neighboring compositions $\mathbf{n}$ and $\mathbf{n} + \mathbf{e}_a - \mathbf{e}_b$. The resulting composition matrix is tridiagonal, so we call these nearest-neighbor entries *Jacobi hopping*.

3 One-pair rigidity of one-body parents

Lemma 3.1 (Separate one-pair zero modes force block preservation)**.** *Let a zero-twist null row be a number-conserving one-body operator $S = d\Gamma(s)$. Suppose*

$$S|\mathbf{e}_b\rangle = 0 \tag{9}$$

for the normalized one-pair AGP in every paired block b. Then

$$P_a s P_b = 0 \qquad (a \neq b) \tag{10}$$

between all paired blocks. If S is Hermitian, the same conclusion includes any unpaired complement coupled to a paired block.

Proof. Fix ordered blocks $a \neq b$, put

$$x = P_a s P_b, \qquad B_x = c_a^\dagger x c_b, \tag{11}$$

and project $S|\mathbf{e}_b\rangle$ onto the charge sector with one fermion in block a and one in block b. That component is exactly $B_x|\mathbf{e}_b\rangle$. The target block is empty in $|\mathbf{e}_b\rangle$. Moreover, $|\mathbf{e}_b\rangle = L_b^{-1/2}\eta_b^+|0\rangle$, and a direct contraction gives

$$\langle \mathbf{e}_b|c_{b,i}^\dagger c_{b,j}|\mathbf{e}_b\rangle = \frac{(J_b J_b^\dagger)_{ji}}{L_b} = \frac{\delta_{ij}}{L_b}, \tag{12}$$

where the last equality is exactly the skew-unitarity of J_b. Therefore

$$\begin{aligned} \|B_x|\mathbf{e}_b\rangle\|^2 &= \langle \mathbf{e}_b|c_b^\dagger x^\dagger c_a c_a^\dagger x c_b|\mathbf{e}_b\rangle \\ &= \frac{1}{L_b}\operatorname{Tr}(x^\dagger x). \end{aligned} \tag{13}$$

The left side vanishes by (9), so $x = 0$. Repeating the argument for every ordered pair proves (10). For a complement without its own pair state, Hermiticity supplies the reverse matrix block. □

Remark 3.2 (Stronger than Hermitian projected Hubbard). The paired-block part of the lemma does not require Hermiticity. If the complete zero manifold contains the separate one-pair AGPs, no additive one-body row in a general $S^\dagger S$ factorization can hide interblock transfer at zero twist. A many-body row can annihilate the same states through a multiplicative null mechanism; a one-body row cannot.

4 Peierls transport of exact interaction subbundles

Let the finite tori be restrictions of fixed translation-covariant covering-lattice kernels. Let $P_a(k)$ be isolated analytic projectors onto interaction singular subbundles. The factor label λ may include a cell, microscopic channel, and positive-kernel factorization index. Assume that the zero-twist one-body kernels obey

$$P_a(k) s_\lambda(k,k') P_b(k') = 0, \qquad a \neq b, \tag{14}$$

for all k, k' on the Brillouin torus.

In a covering-lattice Bloch convention, a complete uniform Peierls connection shifts every momentum argument of every one-body kernel,

$$s_\lambda(A; k, k') = s_\lambda(k + A, k' + A), \tag{15}$$

with fixed orbital-embedding factors included in the symbols. Equation (14), evaluated at $(k+A, k'+A)$, says that the twisted row is block diagonal in the shifted decomposition $P_a(k + A)$.

Because the projectors are isolated and analytic, they admit one transport unitary common to every factor label. To make this explicit, adjoin the orthogonal complement $P_0 = I - \sum_a P_a$ and, along a directional path $A = tv$, write $P_\alpha(t;k) = P_\alpha(k+tv)$. The anti-Hermitian Kato generator

$$\mathcal{K}_v(t;k) = \frac{1}{2}\sum_\alpha [\dot{P}_\alpha(t;k), P_\alpha(t;k)] = \sum_\alpha \dot{P}_\alpha(t;k)P_\alpha(t;k) \tag{16}$$

satisfies $[\mathcal{K}_v, P_\alpha] = \dot{P}_\alpha$. Hence the fiberwise solution

$$\dot{R}_v(t;k) = \mathcal{K}_v(t;k)R_v(t;k), \qquad R_v(0;k) = I, \tag{17}$$

obeys

$$R_v(t;k)P_\alpha(k)R_v(t;k)^\dagger = P_\alpha(k+tv) \tag{18}$$

for the entire mutually orthogonal projector family [4, Chap. II]. The dependence on k is suppressed below. The crucial point is that R_v depends only on the common block decomposition, not on λ. Let $\widehat{R}_v(t)$ be its number-conserving Fock lift.

5 Sharp one-body Peierls-transport no-go theorem

Theorem 5.1 (Peierls-transport obstruction)**.** *Assume:*

- (H1) *$H(A)$ has the analytic positive-square form* (4)*;*
- (H2) *every $S_\lambda(A)$ is a number-conserving one-body row;*
- (H3) *the same zero-twist factor family annihilates every separate one-pair state $|\mathbf{e}_a\rangle$, and at total pair number n its complete kernel is the product-AGP span* (3)*;*
- (H4) *the interaction singular subbundles are isolated and analytic; and*
- (H5) *the physical field dependence is the complete covering-lattice Peierls shift of fixed finite-range or uniformly exponentially local kernels.*

Then the exact curvature map preserves every block composition. Equivalently,

$$\boxed{(Q_v^{(n)})_{\mathbf{mn}} = 0 \quad \textit{for every } \mathbf{m} \neq \mathbf{n}.} \tag{19}$$

In the reduced multi-block notation,

$$\boxed{k_v^{a\leftarrow b} = 0, \qquad \omega_v^{ab} = 0 \quad (a \neq b).} \tag{20}$$

Thus a physical Peierls twist may generate distinct nonzero block masses, diagonal block curvatures, and a nontrivial lower envelope over compositions, but it cannot generate Jacobi hopping between exact one-body interaction blocks.

Proof. By Theorem 3.1, every zero-twist row preserves the exact block decomposition. Under the complete Peierls shift it preserves the shifted decomposition. Pull the shifted family back with the common Kato unitary:

$$S_\lambda(tv) = \widehat{R}_v(t)\widetilde{S}_\lambda(tv)\widehat{R}_v(t)^\dagger, \tag{21}$$

where each $\widetilde{S}_\lambda(tv)$ preserves the fixed charges N_a.

Write

$$G_v = \partial_t \widehat{R}_v(t)|_{t=0}, \qquad \widetilde{B}^{\mathrm{bp}}_{\lambda v} = \partial_t \widetilde{S}_\lambda(tv)|_{t=0}. \tag{22}$$

Differentiating (21) gives

$$B_{\lambda v} = [G_v, S_\lambda] + \widetilde{B}^{\mathrm{bp}}_{\lambda v}. \tag{23}$$

For any $|\psi\rangle \in \mathcal{Z}_n$,

$$[G_v, S_\lambda]|\psi\rangle = -S_\lambda G_v|\psi\rangle. \tag{24}$$

After stacking the factor labels,

$$\big([G_v, S_\lambda]|\psi\rangle\big)_\lambda = -D(0)G_v|\psi\rangle \in \operatorname{ran} D(0). \tag{25}$$

The least-squares projector $\Pi = P_{\ker D(0)^\dagger}$ annihilates this entire common-transport source. Hence

$$T_v = \Pi\big(\widetilde{B}^{\mathrm{bp}}_{\lambda v}\big)_\lambda\big|_{\mathcal{Z}_n}. \tag{26}$$

Every surviving row commutes with every fixed block charge. The zero-twist map $D(0)$ therefore intertwines the block-charge action on Fock space with the diagonal action on its row target. Consequently $\operatorname{ran} D(0)$, $\ker D(0)^\dagger$, and the projector Π are all invariant under those charges. It follows that $T_v|\mathbf{n}\rangle$ and $T_v|\mathbf{m}\rangle$ lie in orthogonal target charge sectors whenever $\mathbf{n} \neq \mathbf{m}$. Their inner product is zero, proving (19). Equation (20) is the same statement expressed through the projected transfer seeds. □

Remark 5.2 (Factor-gauge invariance). An analytic unitary rotation among factor labels contributes a term proportional to $D(0)|\psi\rangle$, which vanishes on the zero manifold. The theorem is therefore independent of the chosen square root of a positive interaction kernel.

Remark 5.3 (The blocks need not be energy bands). No normal-state spectral projector occurs in the proof. The $P_a(k)$ may be eigenprojectors of an interaction Gram operator or any other isolated interaction-defined singular decomposition. Forcing them to coincide with energy bands is unnecessary.

6 Locality and the finite-torus loophole

On one isolated finite torus, an off-block kernel can vanish only on the sampled momentum grid because of an alias or circumference-winding cancellation. A continuous shift of those samples can reveal a derivative.

That is not a fixed-local thermodynamic realization. For a common finite-range or uniformly exponentially local infinite-lattice kernel, the matrix-valued functions

$$F^{ab}_\lambda(k, k') = P_a(k)s_\lambda(k, k')P_b(k') \tag{27}$$

are continuous, and in the exponential case analytic in a strip. If they vanish on the momentum grids of an unbounded regular sequence of tori, they vanish on a dense subset of the Brillouin torus and hence vanish identically. The full kernel identity (14) follows.

Therefore the only finite-torus evasion is nonuniform: its displacement support or Fourier harmonic must scale with the circumference. This is a winding/lift obstruction rather than a local one-body counterexample.

7 Minimal algebraic escape

Theorem 3.1 identifies what must change. An additive one-body row cannot contain interblock transfer while annihilating both separate one-pair AGPs. The smallest structural replacement is a multiplicative null row

$$K = \Pi_{\mathrm{null}}B, \tag{28}$$

where B is an interblock transfer bilinear, $\Pi_{\rm null}$ is a local many-body projector annihilating the zero manifold, and $[\Pi_{\rm null}, B] = 0$. The transfer is present in the operator but hidden at zero twist behind a null projector. This mechanism is impossible for an additive one-body row and is exactly where Theorem 5.1 stops.

The construction below is minimal at the level of theorem hypotheses: it retains translation invariance, strict locality, number conservation, positivity, an auditable complete zero manifold, isolated extensive blocks, and a complete Peierls twist. Only the condition "every null row is one body" is dropped. We do not claim minimality in orbital count or polynomial degree.

8 Strictly local covariant many-body null-row model

Take a periodic chain

$$\Lambda_V = \mathbb{Z}/V\mathbb{Z}, \qquad V \geq 2, \tag{29}$$

with two spinful blocks $a = 1, 2$. Let

$$p^\dagger_{a,x} = c^\dagger_{a,x,\uparrow} c^\dagger_{a,x,\downarrow}, \qquad \eta^+_a = \sum_x p^\dagger_{a,x}. \tag{30}$$

Each block has pair capacity $L_a = V$. The blocks are made inequivalent by choosing distinct positive interaction couplings

$$j_1 \neq j_2 > 0. \tag{31}$$

Thus no block-exchange symmetry identifies their one-pair curvatures; no additional weighted pairing form is needed in the Hamiltonian.

8.1 Local rows

The onsite seniority projector is

$$q_{a,x} = (n_{a,x,\uparrow} - n_{a,x,\downarrow})^2. \tag{32}$$

Let $W_{a,x}$ be the fermionic unitary that swaps the two complete spinful sites (a, x) and $(a, x + 1)$, and define

$$\Pi_{a,x} = \frac{1 - W_{a,x}}{2}. \tag{33}$$

Let

$$W_{12,x} = W_{1,x} W_{2,x}, \qquad \Pi_{12,x} = \frac{1 - W_{12,x}}{2}. \tag{34}$$

Define the crossed interblock transfer

$$X_x = \sum_\sigma \left(c^\dagger_{1,x+1,\sigma} c_{2,x,\sigma} + c^\dagger_{1,x,\sigma} c_{2,x+1,\sigma} \right), \qquad B_x = X_x + X^\dagger_x. \tag{35}$$

The simultaneous swap exchanges the two terms in X_x, so $[\Pi_{12,x}, B_x] = 0$. Hence

$$K_x = \Pi_{12,x} B_x = B_x \Pi_{12,x} \tag{36}$$

is Hermitian, number conserving, and supported on one nearest-neighbor bond.

8.2 Complete bondwise Peierls lift

Orient every covering-lattice bond as $x \to x+1$. Put

$$Y_{a,x} = n_{a,x+1,\uparrow} + n_{a,x+1,\downarrow}, \qquad Y_x = Y_{1,x} + Y_{2,x}. \tag{37}$$

For the boundary bond, $x+1$ means the lifted right endpoint even though its torus label is zero. The accumulated phase around the ring is therefore $\Phi = VA$: this is the standard flux-threading twist, not a globally removable pure gauge. Define

$$\Pi_{a,x}(A) = e^{-\mathrm{i}AY_{a,x}}\Pi_{a,x}e^{\mathrm{i}AY_{a,x}}, \qquad K_x(A) = e^{-\mathrm{i}AY_x}K_x e^{\mathrm{i}AY_x}. \tag{38}$$

Every monomial in the complete swap projector and in the complete product row receives its covering-lattice Peierls phase. Nothing is held fixed while another part is twisted.

The Hamiltonian is

$$\boxed{H_V(A) = \frac{1}{2}\sum_{a,x}(\sqrt{U}\,q_{a,x})^2 + \frac{1}{2}\sum_{a,x}(\sqrt{j_a}\,\Pi_{a,x}(A))^2 + \frac{1}{2}\sum_{x}(gK_x(A))^2,} \tag{39}$$

with $U, j_1, j_2 > 0$ and $g \neq 0$. It is translation invariant and strictly range one, uniformly in V.

9 Complete zero manifold of the escape model

Theorem 9.1 (Exact kernel)**.** *At total particle number $2n$,*

$$\boxed{\ker H_V(0)|_{2n} = \operatorname{span}\{|n_1, n_2\rangle : n_1 + n_2 = n,\ 0 \le n_a \le V\}.} \tag{40}$$

Consequently,

$$\boxed{\dim\ker H_V(0)|_{2n} = \min(V, n) - \max(0, n - V) + 1.} \tag{41}$$

Every odd-particle sector has zero nullity.

Proof. Because the Hamiltonian is a sum of positive squares, a zero state is annihilated by every row. The equations $q_{a,x}|\psi\rangle = 0$ force every local block orbital to be empty or doubly occupied. Thus each block reduces to a hard-core pair configuration.

On this seniority-zero subspace, $W_{a,x}$ acts as the ordinary adjacent transposition of the pair occupations at x and $x+1$. The equations $\Pi_{a,x}|\psi\rangle = 0$ make the state invariant under every adjacent transposition. These transpositions generate the full permutation group S_V. At fixed block filling n_a, its invariant subspace is one dimensional and is spanned by

$$|n_a\rangle_a = \binom{V}{n_a}^{-1/2} \sum_{|A|=n_a} \prod_{x\in A} p^\dagger_{a,x}|0\rangle \propto (\eta_a^+)^{n_a}|0\rangle. \tag{42}$$

Hence the first two row families already have exactly the product-AGP span in (40).

Every product AGP is separately invariant under $W_{1,x}$ and $W_{2,x}$, hence under $W_{12,x}$. Therefore $\Pi_{12,x}|n_1, n_2\rangle = 0$. Using $K_x = B_x\Pi_{12,x}$, every cross row also annihilates every product AGP. Adding the cross squares cannot create new zero modes and removes none of the existing ones. This proves (40). Counting the allowed compositions gives (41). Odd particle number is incompatible with vanishing onsite seniority on every local orbital. ∎

10 Exact non-diagonal composition curvature

Work in the one-pair zero basis

$$|1\rangle = |1,0\rangle, \qquad |2\rangle = |0,1\rangle. \tag{43}$$

Let

$$|a;x\rangle = p^{\dagger}_{a,x}|0\rangle, \qquad |a\rangle = \frac{1}{\sqrt{V}}\sum_{x}|a;x\rangle. \tag{44}$$

10.1 Swap-row sources

For the oriented bond $L = x,\ R = x+1$, define

$$|\phi_{a,x}\rangle = \frac{|a;R\rangle - |a;L\rangle}{\sqrt{2}}. \tag{45}$$

Direct differentiation of (38) gives

$$\partial_A(\sqrt{j_a}\Pi_{a,x}(A))|_0|a\rangle = \mathrm{i}\sqrt{\frac{2j_a}{V}}\,|\phi_{a,x}\rangle, \tag{46}$$

and it annihilates the one-pair state in the other block. Moreover,

$$\sum_x \Pi_{a,x}|\phi_{a,x}\rangle \propto \sum_x(|a;x+1\rangle - |a;x\rangle) = 0. \tag{47}$$

Thus the stacked swap source lies in $\ker D(0)^{\dagger}$ and requires no least-squares subtraction. Its curvature contribution is

$$Q^{(1)}_{\mathrm{swap}} = 2\begin{pmatrix} j_1 & 0 \\ 0 & j_2 \end{pmatrix}. \tag{48}$$

10.2 Cross-row source

Choose normalized split-singlet states $|\chi_x\rangle$ on each bond so that, after a harmless phase convention,

$$\partial_A(gK_x(A))|_0|1\rangle = \frac{2\mathrm{i}g}{\sqrt{V}}|\chi_x\rangle, \qquad \partial_A(gK_x(A))|_0|2\rangle = -\frac{2\mathrm{i}g}{\sqrt{V}}|\chi_x\rangle. \tag{49}$$

One explicit choice is

$$\begin{aligned}|\chi_x\rangle = \frac{1}{2}(&c^{\dagger}_{1,R,\uparrow}c^{\dagger}_{2,L,\downarrow} - c^{\dagger}_{1,R,\downarrow}c^{\dagger}_{2,L,\uparrow} \\ &- c^{\dagger}_{1,L,\uparrow}c^{\dagger}_{2,R,\downarrow} + c^{\dagger}_{1,L,\downarrow}c^{\dagger}_{2,R,\uparrow})|0\rangle.\end{aligned} \tag{50}$$

The local action of K_x on this state is proportional to

$$(|1;R\rangle - |1;L\rangle) - (|2;R\rangle - |2;L\rangle). \tag{51}$$

Summing (51) over the ring telescopes to zero. Hence the stacked cross source also lies in $\ker D(0)^{\dagger}$. Since the factor-row labels are orthogonal in the target, (49) gives

$$Q^{(1)}_{K} = 4g^2\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{52}$$

Theorem 10.1 (Exact one-pair composition operator)**.** *The exact second-order operator is*

$$\boxed{Q^{(1)} = \begin{pmatrix} 2j_1 + 4g^2 & -4g^2 \\ -4g^2 & 2j_2 + 4g^2 \end{pmatrix}.} \tag{53}$$

The sign of the off-diagonal entry changes under a phase reversal of one composition basis vector, but its magnitude does not. Its eigenvalues are

$$\boxed{\lambda_\pm = j_1 + j_2 + 4g^2 \pm \sqrt{(j_1 - j_2)^2 + 16g^4}.} \tag{54}$$

For $j_1, j_2 > 0$ and $g \neq 0$, $\lambda_- > 0$ and $(Q^{(1)})_{12} \neq 0$. Thus

$$E_\pm(A) = \frac{A^2}{2}\lambda_\pm + o(A^2), \tag{55}$$

and the analytic curvature eigenbranches are coherent superpositions of the two block compositions.

Proof. The swap and cross factor labels occupy orthogonal summands of the target space, so their projected Gram matrices add. Equations (48) and (52) give (53). Diagonalization gives (54). Positivity of the lower eigenvalue follows from

$$(j_1 + j_2 + 4g^2)^2 - \left[(j_1 - j_2)^2 + 16g^4\right] = 4j_1j_2 + 8g^2(j_1 + j_2) > 0. \tag{56}$$

□

Remark 10.2 (Extensive-filling scaling)*.* The one-pair hopping $(Q^{(1)})_{12} = -4g^2$ is independent of V, and the finite verifier also finds nonzero off-diagonal response in the $V = 3$, $n = 2$ composition sector. We make no thermodynamic claim here at fixed nonzero pair density n/V. Determining whether an off-diagonal curvature density survives, vanishes, or requires a coupling rescaling in extensive sectors is a separate open problem.

11 Sharpness and interpretation

The theorem and escape model meet at an exact algebraic boundary:

Property	One-body class	Many-body escape
Translation invariant	yes	yes
Strictly or uniformly local	yes	strictly range one
Complete covering-lattice Peierls twist	yes	yes
Two extensive inequivalent interaction blocks	yes	$L_1 = L_2 = V$, $j_1 \neq j_2$
Complete auditable zero manifold	assumed or certified	proved exactly
Positive-square Hamiltonian	yes	yes
Additive one-body null rows only	yes	no
Off-diagonal composition response	impossible	exact and nonzero

The many-body row $K_x = \Pi_{12,x}B_x$ is the minimal structural escape because it combines one null projector with one transfer bilinear. It contains genuine transfer while still annihilating every zero state at $A = 0$. Theorem 3.1 proves that an additive one-body row cannot realize that pattern.

The remaining ways to evade Theorem 5.1 are not equivalent local one-body realizations: a nonuniform circumference-winding family, a non-Peierls field dependence, a singular-value crossing that destroys isolated transport, or abandonment of the complete separate-composition zero manifold.

12 Computational certificates

Two independent deterministic scripts accompany this note. Both now print the composition-basis convention in their headers. The one-body audit uses the ordered basis $(|1,0\rangle, |0,1\rangle)$ and verifies that a common transported off-block source has projected norm below 10^{-15}, that arbitrary block-preserving intrinsic derivatives leave Q diagonal, and that a deliberately noncovariant factor-dependent source produces a large off-diagonal response. The many-body verifier uses the same order, checks the exact kernel in every pair-number sector at $V = 2$ and $V = 3$, verifies (53) through $V = 5$ to errors below 2×10^{-15}, finds nonzero off-diagonal response in a higher-composition sector, and matches direct finite differences. The seeded one-body control and explicit basis headers make the certificate output exactly reproducible.

For

$$U = 2, \qquad j_1 = 1, \qquad j_2 = 1.3, \qquad g = 0.7, \tag{57}$$

one obtains

$$Q^{(1)} = \begin{pmatrix} 3.96 & -1.96 \\ -1.96 & 4.56 \end{pmatrix}, \qquad \operatorname{spec} Q^{(1)} = \{2.27717373, 6.24282627\}. \tag{58}$$

13 Conclusion

The physical two-block program has a sharp answer. Exact local one-body projected-Hubbard or QGN parents with isolated interaction subbundles cannot acquire composition hopping from a complete electromagnetic Peierls twist. Their apparent off-block motion is common subbundle transport and is eliminated by the frustration-free least-squares projection.

A local many-body multiplicative null row is the minimal exact escape in hypothesis space. It can hide interblock transfer at zero twist, retain a completely auditable product-AGP zero manifold, and produce a fully covariant non-diagonal composition-space response. The failed search for a final one-body example is therefore replaced by a structural theorem and by the first exact model beyond its boundary.

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